Resume of Ryan Nathanael Soenjoto Widodo, Ph.D.

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About Me

Hello world! This is a resume page of Ryan Nathanael Soenjoto Widodo. In 1991, Widodo was born in a loving Indonesian Christian family of 4 persons. Widodo received his Ph.D. in 2022 from the University of Tsukuba, Computer Science Department. Currently, Widodo is actively developing Data engineering tools in Rakuten Travel.

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Research interest: Data reduction (compression, deduplication) Storage systems Distributed file systems Non-volatile memory

Research Interest

Widodo has various research interest in Computer Science, including:

[1] Key-value storage
[2] Persistent memory
[3] Big data
[4] Deduplication and compression
[5] Mobile cloud computing

Publications

In chronological order:

[1] Widodo, R. N., Lim, H., & Atiquzzaman, M. (2017). SDM: Smart deduplication for mobile cloud storage. Future Generation Computer Systems, 70, 64-73.
[2] Widodo, R. N., Lim, H., & Atiquzzaman, M. (2017). A new content-defined chunking algorithm for data deduplication in cloud storage. Future Generation Computer Systems, 71, 145-156.
[3] Widodo, R. N., & Lim, H. (2017, January). RDM: Rapid Deduplication for Mobile Cloud Storage. In Proceedings of the 8th International Conference on Computer Modeling and Simulation (pp. 14-18).
[4] Widodo, R. N. S., Abe, H., & Kato, K. (2020, August). HDRF: Hadoop data reduction framework for hadoop distributed file system. In Proceedings of the 11th ACM SIGOPS Asia-Pacific Workshop on Systems (pp. 122-129).
[5] Widodo, R. N. S., Abe, H., & Kato, K. (2020 November). DFS on a Diet: Enabling Reduction Schemes on Distributed File Systems. SC20 Research Poster. Please also check the poster.
[6] Widodo, R. N. S., Abe, H., & Kato, K. (2021). Hadoop Data Reduction Framework: Applying Data Reduction at the DFS Layer. IEEE Access, 9, 152704-152717.
[7] Widodo, R. N. S., Ohtsuji, H., Hayashi, E., Yoshida, E., Abe, H., & Kato, K. (2022). NVKVS: Non-Volatile Memory Optimized Key-Value Separated LSM-Tree. Journal of Information Processing, 30, 332-342.
[8] [Ph.D. Thesis: Design and Implementation of Data Reduction Enabled Distributed File System] (https://tsukuba.repo.nii.ac.jp/record/2005453/files/DA010258.pdf)

For more details, please check my Google Scholar page!

Education

Date	University/Institute	Major	Degree	Comments
August 2009 ~ January 2013	Petra Christian University, Indonesia	Electrical Engineering	Bachelor of Engineering	GPA 3.8/4. Finished in 3.5 years.
February 2015 ~ January 2017	Dongseo Univeristy, South Korea	Ubiquitous IT	Master of Science	Research on mobile cloud storage and deduplication algorithms.
April 2018 ~ March 2019	University of Tsukuba, Japan	Computer Science	Research Student	Research on applying data deduplication in Hadoop.
April 2019 ~ March 2022	University of Tsukuba, Japan	Computer Science	Ph.D. in Computer Science	Research on efficient storage systems for big data.

Work experience

Date	Company	Job description
June 2012 ~ July 2012	PT. Jivan Jaya Makmur Telecom , Surabaya, Indonesia	Managing computer networks with mikrotik routers, June 2012-July 2012
October 2013 ~ September 2014	PT. Hartono Istana Teknologi (Polytron) Kudus, Indonesia	Design and Quality Assurance engineer, responsible for electromagnetic compatibility (EMC) and wireless communications (CISPR 35 and IEC 61000), October 2013-September 2014
April 2022 ~ now	Rakuten	Data Engineer at Rakuten Travel, Data WareHouse (DWH) team. Designing ETL pipeline and developing software for daily batch processings.

Language Proficiency

Language	Level	Proficiency test	Year of test	Comments
Indonesian	Native	-	-	Native language, have been used for more than 20 years actively
English	2nd language	TOEFL IBT (102)	2017	2nd language, used as the primary language since 2015.
Japanese	Intermediate	Not yet	Not planning on taking a test.	Finished level 6 (out of 8) of CEGLOC's Supplementary Japanese course with confidence (grammar A+, kanji A).

Programming Language Proficiency

Language	Year of experience	Comments
Java	~10 years	My go to programming language. Used in many projects including Android applications and Hadoop projects. I can troubleshoot libraries by decompiling the JAR and understanding it from the decompiled codes. I am also well versed with JDBC and ODBC for DB projects.
C++	5 years	The 2nd go to programming language for performance critical project. I have some knowledge on ODBC coding.
C	3 years	Used to program microcontroller, and some additional codes for Java (through JNI interface).
Python	2 years	I built multiple ETL batch jobs using Python with libraries like pyodbc and JayDeBeApi.
C#	1 year	Used for some OpenCV projects and simple Windows interface like GPS display for Embedded Device Class.
Assembly (MCS51)	1 year	Among the first learned programming language. Able to code the microcontroller based on the provided instruction description in the datasheet.
HTML/PHP/Javascript	1 year	Some experience in web development for example, a lab report form interface, SpringBoot projects, and many others.

Data Engineering Proficiency

DB / Tools	Fluency	Experience
RocksDB	Source code fluent	This is a key value store (KVS) developed by Facebook (now Meta). I used this DB in one of my Ph.D. project to demonstrate its performance when used with non-volatile memory (NVM).
Teradata	API/SQL fluent, can troubleshoot issues like NETCONN etc.	This DB is the data warehouse DB for Rakuten Travel, used mainly for analytics. I am well versed for JDBC setup and appication for this DB, including but not limited to: fastload application, tdwallet password extraction for Java application, its SQL, and many others.
Oracle, SQLServer, MySQL, MariaDB, MongoDB	Basic SQL fluent	I have some knowledge on these DB for data extract transform load (ETL).
QlikReplicate	Fluent and can troubleshoot issues	A data copy tool for loading data from one DB to another. I understand most of its API and on how to create and connect DBs with this tool to do table copy jobs and troubleshooting.

DB / Tools	Fluency	Experience
Airflow	Fluent	I use this tool on daily basis (creating and managing DAGs). I can troubleshoot the config and the source code when there is any issue with the tool.
Ansible	Fluent	I wrote several playbook and roles for Rakuten Travel server managements.
Docker	Fluent	Some of the batch jobs that I created are running in containers for data segregation and compatibility with the batch server. I can write Dockerfiles and docker-compose.yml
BigQuery	Self study	I learned BigQuery through resources on Pluralsight. Planning to learn more in the future.

Projects

Name	Status	Collaborator	Comments
An Andorid Application for controlling Quadcopter	Finished	Petrus Santoso (Petra Christian University)	An undergraduate project that involved reverse engineering Parrot AR.Drone 2.0 packets' content and reimplementing it in an Android application. Function includes drone control, status display, and video image display. Please check the project code in GitHub.
Smart Deduplication Mobile	Finished	Professor Hyotaek Lim (Dongseo University), Dr. Mohammed Atiquzzaman (University of Oklahoma)	An Android application aimed toward mobile cloud applications. The proposed application uses deduplication to minimize the storage consumption and data transfer between the user and cloud. It includes an adaptive learning system to select between file deduplication and variable sized-block deduplication depending on the file types.
High throughput Content-Dependent Chunking (CDC) algorithm	Finished	Professor Hyotaek Lim (Dongseo University), Dr. Mohammed Atiquzzaman (University of Oklahoma)	An algorithm that produces variable sized offsets based on the file content. The algorithm is aimed for CDC algorithm used in deduplication applications.
Data reduction schemes in Distributed File Systems	Finished	Professor Kazuhiko Kato (University of Tsukuba), Professor Hirotake Abe (University of Tsukuba)	A project that evaluates the effectivity of data reduction schemes like compressions and deduplications in distributed file systems. The project includes a proposal for including data reduction schemes directly in the DFS layer. Please check the APSys2020 paper for more details.
HDRF's block mirroring	Finished	Professor Kazuhiko Kato (University of Tsukuba), Professor Hirotake Abe (University of Tsukuba)	One of the main issues with HDRF is that any communications between the nodes are using the original data with its original size, wasting network bandwidth and compute because the data will be recompressed again at the receiving node. Because of the modularity of HDRF, we can solve this issue through an additional module called block mirroring. Block mirroring replaces the block replication function of HDFS and transfer the reduced data instead of the original data, minimizing the network usage and compute. Paper:
NVKVS	Finished	Professor Kazuhiko Kato (University of Tsukuba), Professor Hirotake Abe (University of Tsukuba)	An implementation of Wisckey for Intel Optane DCPM memory (Non-Volatile Memory (NVM)). It stores the data in a rotating log in NVM. Project page: https://github.com/NSWRyan/NVKVS
FileWriter	Current		A Java library to make writing to popular formats (CSV, Avro, Parquet, and ORC) easier.
JDBC2JDBC	Current		A Java library to copy data from DB to popular file formats or to another DB via JDBC.
KafkaMySQLDebezium	Current		A self study on change data capture (CDC) technology to make ETL process more efficient.

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